

ICT/0Levels

English Medium

Learn ICT with Shawn sir.

Number Systems

Grade 10

10011001
+ 54A3DF



What are number systems?

A computer scientist says:

- *"Every hexadecimal digit represents exactly four binary digits."*

Using the hexadecimal number **7A3F**₁₆,

prove whether this statement is true by converting the number into binary and explaining the pattern you observe.



A computer stores the binary number: **10110110²**

- Convert it to decimal.
- Convert it to hexadecimal.

Computer Hardware Question--

- A school computer lab experiences slow performance when several applications are opened simultaneously.

Which hardware component would most likely improve this situation when upgraded, and why?

Logic Gates Question.

A security system activates an alarm when:

- A door is open (**D = 1**) AND
- The security system is armed (**A = 1**)

1. Which logic gate represents this situation?
2. Draw its truth table.



Convert the decimal number **173** into:

- a) Binary
- b) Octal
- c) Hexadecimal

Perform the following binary addition:

$$\begin{array}{r} 101101_2 \\ + 110111_2 \\ \hline \end{array}$$

Hexadecimal Conversion

Convert the hexadecimal number **3F7**₁₆ into:

- a) Binary
- b) Decimal

Number System Identification

A student claims that **245**₈ = **245**₁₀

- Is the statement correct?
- If not, convert **245**₈ to decimal and explain the difference.

Binary Subtraction

Perform the following subtraction using binary arithmetic:

$$\begin{array}{r} 1011010_2 \\ - 0110111_2 \\ \hline \end{array}$$



Data Storage Calculation

A computer stores a student ID using **16 bits**.

- What is the largest decimal number that can be represented using 16 bits?
- How many unique student IDs can be represented?

Error Detection

One of the following binary numbers is invalid. Identify it and explain why.

- 101101_2
- 110210_2
- 111000_2
- 100011_2

Binary Logic and Number Systems

Convert 110101_2 to decimal.

Then determine whether the decimal value is:

- Even or odd
- Divisible by 3

Show your working.



Real-World Problem

A security system uses a **4-digit hexadecimal code**.

- How many different codes are possible?
- If a hacker can try one code every second, approximately how many hours would it take to try every possible code?

Real-World argument

Andrew is a computer scientist, and insists the following number is a decimal number and not a binary. Is he right or wrong?, explain your answer.

- **10110101**

Challenge Question

A number is represented as **101101₂**

If the leftmost bit is removed, the new number becomes **01101₂**

- Find the decimal value of both numbers.
- By how much did the value decrease?
- Explain why removing a digit from the left side changes the value much more than removing a digit from the right side.



Real-World Problem

Samantha sold a lollypop for 20 rupees and 50 cents.

- Convert only the integer number into binary
- Now convert the fraction into binary and show the full Rs20.50/- in binary

Show your workings

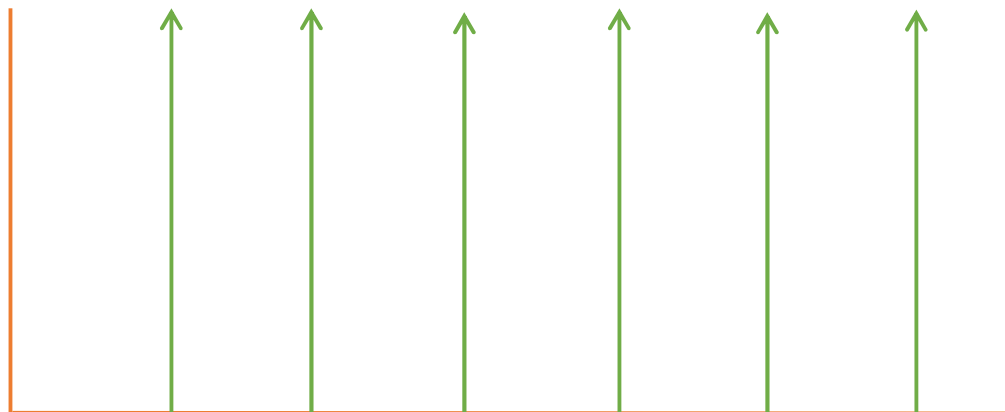
Real-World argument

David added the following two binary fractions and got the correct value. Now show David how to convert this into Decimal.

$$\begin{array}{r}
 101001.1001 \\
 + 1100.1101 \\
 \hline
 110110.0110
 \end{array}$$

Represent the following 4 numbers in the below shown abacus. Use a pencil to draw circles in the right location (you can then erase them and redraw the next number)

987 , 3490 , 1209 , 5600



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